12 THE HALOGENS

The elements in Group 7 of the Periodic Table are called the halogens. In this chapter we will look at some of the properties of these elements. When we first look at them, it is hard to see that they form a family of elements because they look so different, but they react in very similar ways.



▶ Figure 12.1 Chlorine, bromine and iodine are three of the halogens.

LEARNING OBJECTIVES

- Know the colours, physical states (at room temperature) and trends in physical properties of the halogens
- Use knowledge of trends in Group 7 to predict the properties of other halogens
- Understand how displacement reactions involving halogens and halides provide evidence for the trend in reactivity in Group 7

CHEMISTRY ONLY

- Explain the trend in reactivity in Group 7 in terms of electronic configurations.

F	fluorine
Cl	chlorine
Br	bromine
I	iodine
At	astatine

▲ Figure 12.2 The halogens



▲ Figure 12.3 Iodine is a grey solid but has a purple vapour.

THE HALOGENS

The atoms of fluorine, chlorine, bromine, iodine and astatine all have 7 electrons in their outer shell, therefore these elements are put in Group 7 in the Periodic Table. The number of electrons in the outer shell of an atom determines how something reacts, so all the elements in Group 7 react in a similar way.

The name '**halogen**' means 'salt-producing'. When they react with metals, these elements produce a wide range of salts, including calcium fluoride, sodium chloride, silver bromide and potassium iodide. All the salts contain the X^- ion (where X stands for any halogen atom).

The halogens are non-metallic elements with diatomic molecules: F_2, Cl_2, etc.

Table 12.1 The melting and boiling points of the halogens increase and the colour becomes darker down the group

	Physical state at room temperature	Colour
F_2	gas	yellow
Cl_2	gas	green
Br_2	liquid	red-brown liquid, orange/brown vapour
I_2	solid	grey solid, purple vapour

REMINDER

Remember that intermolecular forces are the forces of attraction between molecules.

The melting and boiling points increase down the group. The halogens are all covalent molecular substances and the melting and boiling points increase as the relative molecular mass increases. As the relative molecular mass increases, the intermolecular forces of attraction become stronger and therefore more energy must be put in to overcome these stronger forces of attraction. Remember, no covalent bonds are broken when these melt/boil.

Because the halogens are non-metals, they are poor conductors of heat and electricity.

Astatine is radioactive and is formed during the radioactive decay of other elements, such as uranium and thorium. Most of its isotopes are so unstable that their lives can be measured in seconds or fractions of a second.

SAFETY

Fluorine is so dangerously reactive that you would never expect to find it in a school lab. Apart from any safety problems, due to the reactivity of the elements (especially fluorine and chlorine), all the halogens have extremely poisonous **vapours** and have to be handled in a fume cupboard. Liquid bromine is also very corrosive, and great care has to be taken to keep it off the skin.

REACTIONS OF THE HALOGENS

The halogens react with hydrogen to form **hydrogen halides**: hydrogen fluoride, hydrogen chloride, hydrogen bromide and hydrogen iodide. For example:

$$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$$

KEY POINT

This is something that can be confusing. Hydrogen chloride is a gas and when it dissolves in water it becomes hydrochloric acid. Hydrogen chloride is a gas consisting of HCl molecules and hydrochloric acid is a solution containing H^+ and Cl^- ions. The acids formed by other hydrogen halides are:

HF(aq) hydrofluoric acid
HBr(aq) hydrobromic acid
HI(aq) hydroiodic acid

The hydrogen halides are all acidic, poisonous gases. In common with all the compounds formed between the halogens and non-metals, the gases are covalently bonded. They are very soluble in water, reacting with it to produce solutions of acids. For example, hydrochloric acid is a solution of hydrogen chloride in water:

$$HCl(g) \xrightarrow{\text{dissolve in water}} HCl(aq)$$

hydrogen chloride hydrochloric acid

The halogens react with alkali metals to form salts. For instance, sodium burns in chlorine with its typical yellow flame to produce white, solid sodium chloride:

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

Sodium chloride is an ionic solid. Typically, when the halogens react with metals from Groups 1 and 2, they form ions.



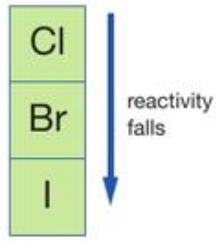
▲ Figure 12.4 Sodium burning in chlorine to produce sodium chloride.

DISPLACEMENT REACTIONS INVOLVING THE HALOGENS

In this section we will look at what happens when we react a solution of a halogen with a solution containing halide ions. These are called **displacement reactions** – you will see why below. We can use these displacement reactions to show that chlorine is more reactive than bromine, which is more reactive than iodine.

KEY POINT

We will concentrate on the three commonly used halogens, but the trend continues for the rest of the group.



▲ Figure 12.5 Reactivity decreases down the group.

REACTING CHLORINE WITH POTASSIUM BROMIDE OR POTASSIUM IODIDE SOLUTIONS

KEY POINT

Although pure liquid bromine is red-brown, in solution it is orange. If the solution is very dilute it may even look yellow.

HINT

You have to be careful with the words you use here. In the exam you must write 'chlorine displaces bromine' not 'chlorine displaces bromide'.



▲ Figure 12.6 Bromine and iodine displaced from potassium bromide and potassium iodide solutions.

If you add chlorine solution ('chlorine water') to colourless potassium bromide solution, the solution becomes orange as bromine is formed:

$$2KBr(aq) + Cl_2(aq) \rightarrow 2KCl(aq) + Br_2(aq)$$

potassium bromide + chlorine → potassium chloride + bromine

Chlorine is more reactive than bromine and has displaced the bromine from solution. You can think about this in the following way: if something is more reactive, it has a greater tendency to *react* to form a compound. Something that is less reactive is more likely to go back to being the element (the unreacted form).

Similarly, adding chlorine solution to potassium iodide solution gives a brown (orange if it is very dilute) solution of iodine.

$$2KI(aq) + Cl_2(aq) \rightarrow 2KCl(aq) + I_2(aq)$$

In exactly the same way, the more reactive bromine displaces the less reactive iodine from potassium iodide solution. Adding bromine solution ('bromine water') to colourless potassium iodide solution gives a brown solution of iodine:

$$2KI(aq) + Br_2(aq) \rightarrow 2KBr(aq) + I_2(aq)$$



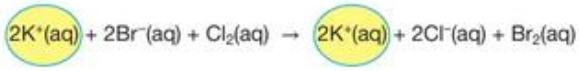
▲ Figure 12.7 Bromine displaces iodine from potassium iodide solution.

All of the following experiments show no reaction:

- bromine + potassium chloride solution: the less reactive bromine cannot displace the more reactive chlorine
- iodine + potassium chloride solution: the less reactive iodine cannot displace the more reactive chlorine
- iodine + potassium bromide solution: the less reactive iodine cannot displace the more reactive bromine.

IONIC EQUATIONS FOR DISPLACEMENT REACTIONS

In all of the above reactions it would not have mattered if we had reacted chlorine with *sodium* bromide or *magnesium* bromide, for example. The reaction occurs between the chlorine molecule and the bromide ion; the metal ion does not do anything, we call it a **spectator ion**. If we write the reaction between chlorine and potassium iodide showing all the charges, we can see this more clearly:



You can see that the potassium ions are the same on each side. They have not changed and so we can write a new equation simply concentrating on the things that have changed:

$$2Br^-(aq) + Cl_2(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$$

This is called an *ionic equation* and does not include the spectator ions.

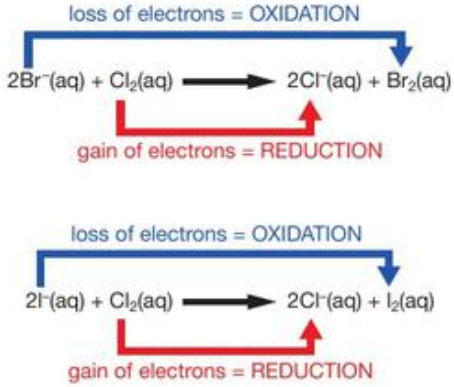
REMINDER

Remember when an ionic compound dissolves in water, it splits into its ions.

DISPLACEMENT REACTIONS AS REDOX REACTIONS

The reactions above that do show a colour change are often discussed in terms of oxidation and reduction, they are *redox reactions*. If you have not already covered redox reactions, you should look at Chapter 14 before reading the next section.

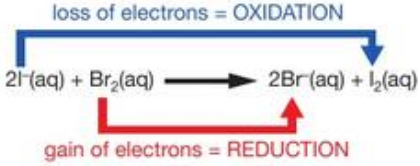
If we look at the reaction between chlorine and either bromide ions or iodide ions, we can see why they are described as redox reactions.



▲ Figure 12.8 The reaction of chlorine with potassium bromide (top) or potassium iodide (below).

In each case, the chlorine is acting as an oxidising agent. In the first equation it oxidises the Br^- ions by taking electrons away from them. Because the oxidising agent takes electrons away from something else, it is reduced itself: the Cl_2 is reduced to $2Cl^-$ ions.

In the reaction between bromine and iodide ions, the bromine is the oxidising agent:



▲ Figure 12.9 The reaction between bromine solution and potassium iodide solution.

REMINDER

Some reminders about oxidation and reduction:
oxidation is loss of electrons
reduction is gain of electrons
An oxidising agent is something that oxidises something else by taking electrons away from it. Oxidising agents accept electrons and therefore are the things that are reduced in a reaction.
A reducing agent is something that reduces something else by giving electrons to it. Reducing agents give away electrons and therefore are the things that are oxidised in a reaction.

EXTENSION WORK

This only works for non-metals. For metals, the most reactive one is the best reducing agent. The difference comes because non-metals react to form negative ions but metals react to form positive ions.

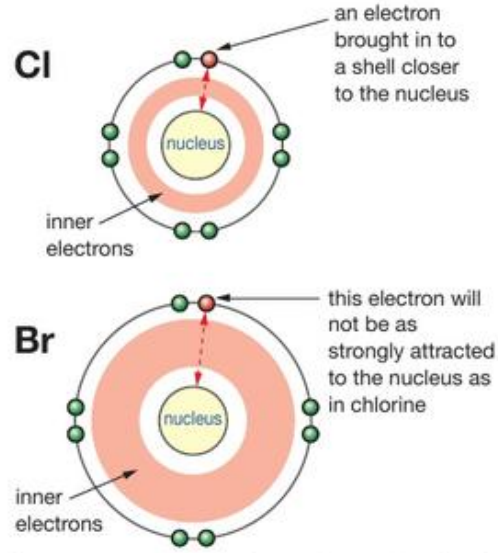
EXTENSION WORK

We can now say a bit more than just 'chlorine is more reactive than bromine, which is more reactive than iodine'. We can describe the reactivity in terms of how good an oxidising agent something is: chlorine is a stronger oxidising agent than bromine, which is a stronger oxidising agent than iodine. Chlorine can oxidise (take electrons away from) bromide ions and iodide ions; bromine can oxidise (take electrons away from) iodide ions, but not chloride ions; iodine cannot oxidise bromide ions or chloride ions.

CHEMISTRY ONLY

EXPLAINING THE TREND IN THE REACTIVITY OF THE HALOGENS

We can explain the reactivity of the halogens in terms of how readily they form negative ions. When the halogens react (in these reactions) they form 1– ions by taking electrons away from something else. A chlorine atom is smaller than a bromine atom, so when we add an electron to the outer shell of a chlorine atom, we are adding it to a shell closer to the nucleus. If it is in a shell closer to the nucleus it is more strongly attracted to the nucleus. The bromine atom is bigger and so the outer electron is added to a shell further away from the nucleus, where it is not as strongly attracted. Chlorine therefore has a stronger tendency to form a 1– ion than bromine, and a chlorine atom will take an electron away from a bromide ion.



▲ Figure 12.10 Chlorine has a stronger attraction for an electron than bromine.

EXTENSION WORK

This argument does not apply to fluorine. Fluorine atoms don't, in fact, accept electrons more easily than chlorine atoms do. This is for reasons which we don't study at International GCSE. The greater reactivity of fluorine is because of a particularly weak covalent bond between the fluorine atoms. In the exam, you should not be asked to explain why fluorine is more reactive than chlorine, but, if you are, you will have to use the explanation described here, even though it is wrong!

EXTENSION WORK

You may be thinking why doesn't the higher nuclear charge on a bromine atom mean that it attracts electrons more strongly? You have to consider the amount of attraction the incoming electron actually feels from the nucleus. In chlorine, there are 17 positively charged protons balanced by the 10 negatively charged electrons in the inner shells. That means the new electron feels an overall pull of 7+ from the centre of the atom. A similar argument with bromine shows that the new electron also feels an overall pull from the nucleus of 7+, but in the bromine case, it is further away.

This explanation will not be required at International GCSE.

END OF CHEMISTRY ONLY

SUMMARISING THE MAIN FEATURES OF THE GROUP 7 ELEMENTS

Group 7 elements:

- have diatomic molecules, X_2 (e.g. F_2, Cl_2, Br_2)
- go from gases to liquid to solid as you move down the group: the melting points and boiling points increase down the group
- have coloured poisonous vapours: the colours of the elements get darker down the group
- form compounds with the formula HX (e.g. HF, HCl, HBr) when reacted with hydrogen and these dissolve in water to form acids
- form ionic salts with metals and covalent compounds with non-metals
- form X^- (e.g. F^-, Cl^-, Br^-) ions in ionic compounds
- become less reactive towards the bottom of the group
- will displace elements lower down the group from their salts.

We can use these properties and trends to predict the properties of astatine, the element below iodine in Group 7. Astatine is an extremely rare radioactive element and no one has ever seen a sample of it, but we can predict that it:

- will be a darker colour than iodine, very dark grey or black
- will be a solid at room temperature and have a higher melting point than iodine
- will be diatomic and contain At_2 molecules
- will react with hydrogen to form HAt, which will dissolve in water to form an acid
- will form the salt NaAt with sodium or KAt with potassium
- will contain the astatide ion (At^-) in its ionic salts
- will be less reactive than iodine
- will be displaced from solution by iodine

CHAPTER QUESTIONS

SKILLS INTERPRETATION 8

1 a Draw a diagram to show the arrangement of electrons in a fluorine atom.

SKILLS REASONING 7

b Explain why fluorine is placed in Group 7 in the Periodic Table.

SKILLS INTERPRETATION 8

c Draw a dot-and-cross diagram (showing outer electrons only) of the molecule formed when fluorine reacts with hydrogen.

SKILLS REASONING 7

d Explain why fluorine has a lower melting point than chlorine.

e Explain what you would observe when a chlorine solution is added to a solution of potassium fluoride.

SKILLS INTERPRETATION (8)

2 This question is about astatine, At, at the bottom of Group 7 of the Periodic Table. You are asked to make some predictions about astatine and its chemistry.

a Showing only the outer electrons, draw diagrams to show the arrangement of electrons in an astatine *atom*, an astatide *ion* and an astatine *molecule*.

SKILLS REASONING (7)

b Explain what physical state you would expect astatine to be in at room temperature.

c Explain whether you would expect astatine to be more or less reactive than iodine.

(6)

d Suggest a likely pH for a reasonably concentrated solution of hydrogen astatide in water.

(8)

e Explain what you would expect caesium astatide to look like. Will it be soluble in water? Explain your reasoning.

SKILLS PROBLEM SOLVING, REASONING (10)

f Write an ionic equation for the reaction that will occur if you add chlorine water to a solution of sodium astatide. Assume that astatine is insoluble in water. Explain clearly why this reaction would be counted as a redox reaction.

SKILLS PROBLEM SOLVING (9)

3 Chlorine reacts with fluorine to form chlorine monofluoride (ClF).

a Write an equation for this reaction, including state symbols. All the substances are gases at room temperature.

SKILLS INTERPRETATION (8)

b Draw a dot-and-cross diagram for ClF.

SKILLS REASONING

c Arrange the substances Cl_2, F_2 and ClF in order of increasing boiling point and explain your order.

SKILLS PROBLEM SOLVING

d Chlorine and fluorine can also react together to form chlorine trifluoride, which is a gas at room temperature and pressure.

i State the formula of chlorine trifluoride.

(9)

ii Chlorine trifluoride is an extremely reactive substance and, for instance, reacts explosively with water. The products of the reaction are hydrofluoric acid, hydrochloric acid and oxygen gas. Write an equation, including state symbols, for the reaction of chlorine trifluoride with water.